

AAEC 4484/AAEC(STAT) 5484: Applied Economic Forecasting

Alex Fuentes

Homework #7 - Spring 2026

Instructions: This question is a modified version of the Empirical example in Ghysels & Marcellino (2018) – Applied economic forecasting using time series methods, Ch. 6.

A PDF version of the question and expected output is available in the HW7 folder. **Caution: The questions and results might not be a 1-to-1 mapping with the textbook, so be sure to read carefully and answer the specific prompts below.**

Question 1: Euro Area Business Cycles

Mawuena is a research economist at the European Central Bank tasked with understanding how macroeconomic shocks transmit across the major Euro area economies. He has assembled a dataset of quarterly real GDP growth rates (year-over-year, percent) for France (`g_fr`), Germany (`g_ger`), and Italy (`g_it`), along with the United States (`g_us`), spanning 1986Q1 to 2012Q4. The dataset also contains some dummy variables labeled `d_XXXXqx` accounting for exceptional (outlier) episodes across the sample. The data are stored in `euro_growth.csv` on our GitHub Data folder.

Before diving in, we need to address a modeling choice that is not explicitly covered in our slides but will come up constantly in your own work: **exogenous variables in a VAR**. The standard `vars::VAR()` function accepts an **exogen** argument— a matrix of variables that enter every equation as contemporaneous controls but whose own dynamics are *not* modeled by the system. The United States is a natural candidate here. It is large enough that French, German, or Italian business cycles are unlikely to move the US economy in any meaningful way, but a US recession clearly spills over into Europe. If we had treated `g_us` as endogenous, we would force the model to “explain” it using European lags, wasting parameters and potentially distorting the dynamics we actually care about. Treating it as exogenous instead lets it absorb common global shocks in every equation while keeping the system focused on the inter-European dynamics we want to study.

Unless otherwise specified, `g_us` and the dummy variables should be included as exogenous controls in all VAR estimations below.

Data and Stationarity

- (a) Import the `euro_growth.csv` file and convert to a `tsibble` object with a `yearquarter` index. Plot the three EU growth series on a single graph. To motivate the idea of US growth-induced comovements among the EU series, please super impose the US recessions (as identified by the NBER¹) over the sample periods. **Remember that ordering of the `ggplot` layers matter so make the recession bars appear as shaded areas behind the lines. Hence, this needs to be the base plot.**

Briefly describe what you see — do the three European series move together? Are there any episodes that stand out?

¹You are interested in the Peak and Trough dates here: <https://www.nber.org/research/data/us-business-cycle-expansions-and-contractions>

They do seem to move together quite consistently throughout the plot. Germany seems to have the highest lows and peaks. You can also see multiple areas of recession. A strange recession area where France and Italy both but Germany shoots up drastically. This could be due to the German reunification though. A dramatic drop in the 2008-2009 range of time is also very obvious.

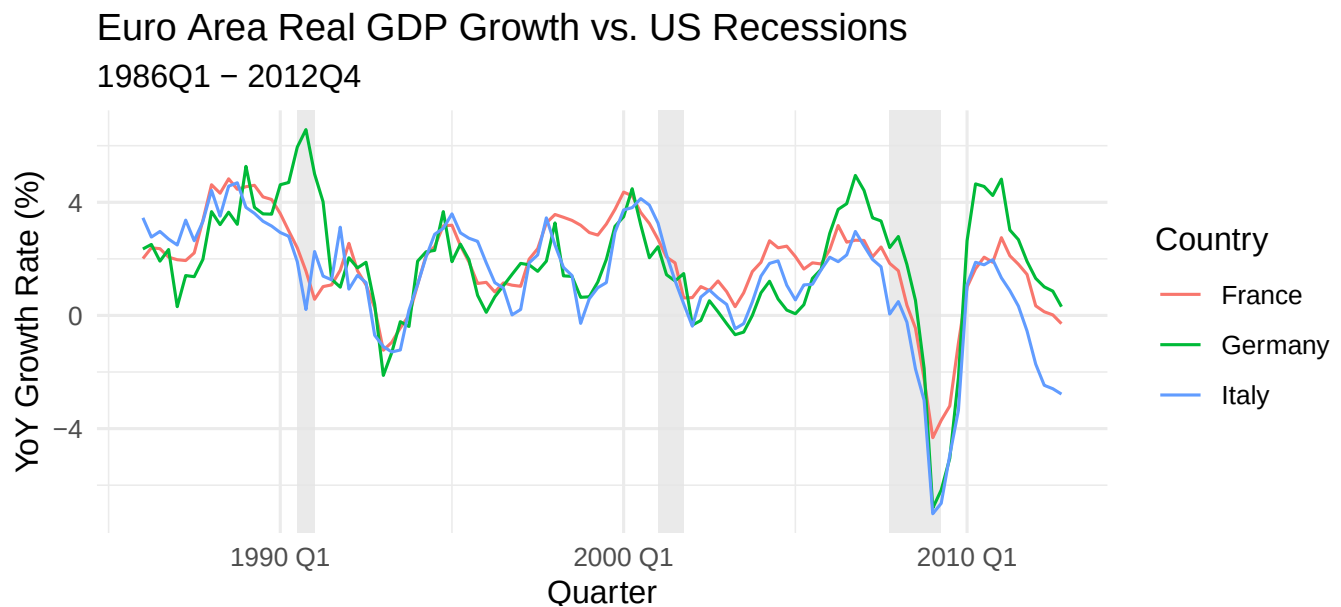
```
url <- "https://raw.githubusercontent.com/Shamar-Stewart/Forecasting-S26/refs/heads/main/Data/eu_growth.csv"

euro_growth_raw <- read_csv(url)

euro_growth <- euro_growth_raw %>%
mutate(time = yearquarter(str_replace(date, ":", " Q")) %>%
as_tsibble(index = time)

recessions <- data.frame(
start = yearquarter(c("1990 Q3", "2001 Q1", "2007 Q4")),
end   = yearquarter(c("1991 Q1", "2001 Q4", "2009 Q2")))

ggplot() +
geom_rect(data = recessions,
aes(xmin = start, xmax = end, ymin = -Inf, ymax = Inf),
fill = "grey90", alpha = 0.8) +
geom_line(data = euro_growth, aes(x = time, y = g_fr, color = "France")) +
geom_line(data = euro_growth, aes(x = time, y = g_ger, color = "Germany")) +
geom_line(data = euro_growth, aes(x = time, y = g_it, color = "Italy")) +
labs(title = "Euro Area Real GDP Growth vs. US Recessions",
subtitle = "1986Q1 - 2012Q4",
x = "Quarter", y = "YoY Growth Rate (%)", color = "Country") + theme_minimal()
```



- (b) Apply the ADF test (`ur.df()`) to `g_fr`, `g_ger`, and `g_it`. For each series, justify your choice of test specification ("drift" vs. "trend") and comment on your conclusions at the 5% level of significance?

Hint: Use a lag length of 4 for the ADF test, which is a common choice for quarterly data.

Decided to use "drift" for all of the test due to the expectation of the mean being non-zero and the plot showing no real trend pattern to test on, at least from my discernment.

```
 #(test statistic)
```

adf_fr| -2.3991|We fail to reject the null hypothesis of this being a non-stationary process and conclude that it is non-stationary. adf_ger| -3.4502|We reject the null hypothesis of this being a non-stationary process leading to the conclusion of stationarity. adf_it| -2.2048| We fail to reject the null hypothesis of this being a non-stationary process and conclude this is non-stationary.

```
adf_fr <- ur.df(euro_growth$g_fr, type = "drift", lags = 4)
adf_ger <- ur.df(euro_growth$g_ger, type = "drift", lags = 4)
adf_it <- ur.df(euro_growth$g_it, type = "drift", lags = 4)
```

```
summary(adf_fr)
```

```
##
## #####
## # Augmented Dickey-Fuller Test Unit Root Test #
## #####
##
## Test regression drift
##
##
## Call:
## lm(formula = z.diff ~ z.lag.1 + 1 + z.diff.lag)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.587 -0.300  0.038  0.355  1.529
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    0.1706     0.0928   1.84   0.0690 .
## z.lag.1        -0.0981     0.0409  -2.40   0.0183 *
## z.diff.lag1     0.3666     0.0876   4.19  6.2e-05 ***
## z.diff.lag2     0.2654     0.0951   2.79   0.0064 **
## z.diff.lag3     0.1217     0.0977   1.25   0.2157
## z.diff.lag4    -0.3874     0.0952  -4.07  9.6e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.542 on 97 degrees of freedom
## Multiple R-squared:  0.43,    Adjusted R-squared:  0.401
## F-statistic: 14.7 on 5 and 97 DF,  p-value: 1.09e-10
##
##
## Value of test-statistic is: -2.4 2.9
##
## Critical values for test statistics:
##      1pct  5pct 10pct
## tau2 -3.46 -2.88 -2.57
## phi1  6.52  4.63  3.81
```

```
summary(adf_ger)
```

```
##
## #####
```

```
## # Augmented Dickey-Fuller Test Unit Root Test #
## #####
##
## Test regression drift
##
##
## Call:
## lm(formula = z.diff ~ z.lag.1 + 1 + z.diff.lag)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -4.567 -0.473 -0.043  0.615  2.834
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   0.4000     0.1547   2.59 0.01122 *
## z.lag.1      -0.2253     0.0653  -3.45 0.00083 ***
## z.diff.lag1   0.2896     0.0902   3.21 0.00180 **
## z.diff.lag2   0.2304     0.0939   2.45 0.01590 *
## z.diff.lag3   0.1198     0.0967   1.24 0.21841
## z.diff.lag4  -0.2252     0.0974  -2.31 0.02282 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.05 on 97 degrees of freedom
## Multiple R-squared:  0.302, Adjusted R-squared:  0.266
## F-statistic: 8.38 on 5 and 97 DF,  p-value: 1.29e-06
##
##
## Value of test-statistic is: -3.45 5.95
##
## Critical values for test statistics:
##      1pct  5pct 10pct
## tau2 -3.46 -2.88 -2.57
## phi1  6.52  4.63  3.81
```

```
summary(adf_it)
```

```
##
## #####
## # Augmented Dickey-Fuller Test Unit Root Test #
## #####
##
## Test regression drift
##
##
## Call:
## lm(formula = z.diff ~ z.lag.1 + 1 + z.diff.lag)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.404 -0.459  0.071  0.500  2.503
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   0.1003     0.1040   0.96 0.3371
```

```
## z.lag.1      -0.1102      0.0500     -2.20     0.0298 *
## z.diff.lag1   0.2578      0.0802      3.21     0.0018 **
## z.diff.lag2   0.2002      0.0831      2.41     0.0179 *
## z.diff.lag3   0.2620      0.0853      3.07     0.0028 **
## z.diff.lag4  -0.5218      0.0880     -5.93    4.7e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.808 on 97 degrees of freedom
## Multiple R-squared:  0.431, Adjusted R-squared:  0.401
## F-statistic: 14.7 on 5 and 97 DF, p-value: 1.05e-10
##
##
## Value of test-statistic is: -2.2 2.6
##
## Critical values for test statistics:
##      1pct  5pct 10pct
## tau2 -3.46 -2.88 -2.57
## phi1  6.52  4.63  3.81
```

Lag Selection and Estimation

Although the ADF test might have failed to reject the null of a unit root for some of the series above, this likely reflects the well-known low power of unit root tests in small samples and in the presence of the potential structural breaks identified. From an economic standpoint, GDP growth rates are generally stationary processes. Therefore, we proceed under the assumption of stationarity.

- (c) Next, we will proceed by using the data from 1986 Q1 – 2003 Q4 for the model estimation and 2004 Q1 – 2012 Q4 for the diagnostics.

Use `vars::VARselect()` with a `lag.max = 4` to select the lag order, passing `g_fr`, `g_ger`, and `g_it` as the endogenous block and `g_us` and the recession dummies (from the `.csv` file) as the exogenous variable.

Go through the process of testing to ensure that the lag order(s) suggested has (have) correctly specified the dynamics of the system.

Note: In the `serial.test()`, use `lags.pt = 16` and the 5% level of significance to check for autocorrelation up to 4 years. If the diagnostics suggest that you need to add more lags, do so and re-run the diagnostics until you find a lag order that eliminates autocorrelation in the residuals.

Using the null hypothesis of the process being white noise and an alternative hypothesis of the process not being white noise thus concluding we could extract more information. The Portmanteau Test at the $p=2$ level returns a p-value of .1 which is higher than 0.05 and we fail to reject the null hypothesis. Concluding that the process is stationary and captures the dynamics sufficiently. The test at the $p=1$ level yielded a p-value of .004 much lower than .05 leads us to select the model at $p=2$. We fail to reject the null hypothesis and deem it insufficient to use further on.

```
train_data <- euro_growth %>% filter(time <= yearquarter("2003 Q4"))

y_train <- train_data %>%
  as.data.frame() %>%
  dplyr::select(g_fr, g_ger, g_it)

exog_train <- train_data %>%
  as.data.frame() %>%
  dplyr::select(g_us, starts_with("d_"))
```

```
lag_selection <- VARselect(y_train, lag.max = 4, type = "const", exogen = exog_train)
print(lag_selection$selection)
```

```
## AIC(n)  HQ(n)  SC(n) FPE(n)
##      2      2      1      2
```

```
var_model <- VAR(y_train, p = 1, type = "const", exogen = exog_train)
```

```
serial_diag <- serial.test(var_model, lags.pt = 16, type = "PT.asymptotic")
print(serial_diag)
```

```
##
## Portmanteau Test (asymptotic)
##
## data: Residuals of VAR object var_model
## Chi-squared = 183, df = 135, p-value = 0.004
```

```
## $serial
##
## Portmanteau Test (asymptotic)
##
## data: Residuals of VAR object var_model
## Chi-squared = 183, df = 135, p-value = 0.004
```

```
var_model_p2 <- VAR(y_train, p = 2, type = "const", exogen = exog_train)
```

```
serial.test(var_model_p2, lags.pt = 16, type = "PT.asymptotic")
```

```
##
## Portmanteau Test (asymptotic)
##
## data: Residuals of VAR object var_model_p2
## Chi-squared = 147, df = 126, p-value = 0.1
```

- (d) Estimate your chosen VAR over the subsample. Report the coefficient table for the **Germany equation only**. Is the US growth variable significant in this equation? Does this match your economic intuition? **Remember your exogenous variables!**

The US growth variable is not significant in this equation, It does not seem to follow economic intuition as you would think there would be a link between the two but external and internal country politics could be playing a role here. Plus the US could have a delayed effect on Germany so with it being exogenous it only is reviewed for the present impact.

```
summary(var_model_p2)$varresult$g_ger
```

```
##
## Call:
## lm(formula = y ~ -1 + ., data = datamat)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.6534 -0.4698 -0.0179  0.4727  2.0467
```

```
##
## Coefficients:
##      Estimate Std. Error t value Pr(>|t|)
## g_fr.l1      0.0504     0.2757   0.18  0.8555
## g_ger.l1      0.7248     0.1235   5.87 2.1e-07 ***
## g_it.l1       0.2706     0.1485   1.82  0.0734 .
## g_fr.l2       0.0960     0.2345   0.41  0.6837
## g_ger.l2      0.0976     0.1320   0.74  0.4624
## g_it.l2     -0.3620     0.1591  -2.27  0.0266 *
## const       -0.2338     0.3633  -0.64  0.5224
## g_us         0.1384     0.1177   1.18  0.2443
## d_1988q1     1.0638     0.6555   1.62  0.1100
## d_1993q1    -1.9162     0.6932  -2.76  0.0076 **
## d_2000q1     0.2834     0.6737   0.42  0.6755
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.866 on 59 degrees of freedom
## Multiple R-squared:  0.787, Adjusted R-squared:  0.75
## F-statistic: 21.7 on 10 and 59 DF,  p-value: 2.81e-16
```

- (e) Run the remaining suite of diagnostics on your estimated VAR: normality (`normality.test()`), heteroskedasticity (`arch.test()`), and stability check (`vars::roots()`). Summarize your findings in a short paragraph. Be sure to mention the null hypotheses and your conclusions in each case. If any diagnostic raises a concern, explain what you would do about it.

The VAR2 model passes all of the diagnostic checks, proving it is a reliable choice. The normality test (null hypothesis: residuals are normal) has a p-value of 0.2, so we fail to reject the null and conclude the errors are normally distributed. The ARCH test (null hypothesis: have constant variance) returns a p-value of 0.5, thus we fail to reject the null hypothesis and find no issues with changing volatility. Finally, the stability check tests if all roots are below 1 and our highest maximum was 0.818, which confirms the system is stable and shocks will eventually die out. Since all tests passed using VAR2, there are no concerns. If normality had not passed, I could have have used bootstrapping for more robust confidence intervals, but that is not necessary given the results.

```
norm_test <- normality.test(var_model_p2)
print(norm_test)
```

```
## $JB
##
## JB-Test (multivariate)
##
## data: Residuals of VAR object var_model_p2
## Chi-squared = 8, df = 6, p-value = 0.2
##
##
## $Skewness
##
## Skewness only (multivariate)
##
## data: Residuals of VAR object var_model_p2
## Chi-squared = 5, df = 3, p-value = 0.1
##
##
## $Kurtosis
##
```

```

## Kurtosis only (multivariate)
##
## data: Residuals of VAR object var_model_p2
## Chi-squared = 3, df = 3, p-value = 0.5

## $jb.mul
## $jb.mul$JB
##
## JB-Test (multivariate)
##
## data: Residuals of VAR object var_model_p2
## Chi-squared = 8, df = 6, p-value = 0.2
##
##
## $jb.mul$Skewness
##
## Skewness only (multivariate)
##
## data: Residuals of VAR object var_model_p2
## Chi-squared = 5, df = 3, p-value = 0.1
##
##
## $jb.mul$Kurtosis
##
## Kurtosis only (multivariate)
##
## data: Residuals of VAR object var_model_p2
## Chi-squared = 3, df = 3, p-value = 0.5

arch_test <- arch.test(var_model_p2, lags.multi = 5)
print(arch_test)

##
## ARCH (multivariate)
##
## data: Residuals of VAR object var_model_p2
## Chi-squared = 180, df = 180, p-value = 0.5

## $arch.mul
##
## ARCH (multivariate)
##
## data: Residuals of VAR object var_model_p2
## Chi-squared = 180, df = 180, p-value = 0.5

var_roots <- roots(var_model_p2)
print(var_roots)

## [1] 0.818 0.818 0.522 0.522 0.312 0.312

```

Part III: Granger Causality

- (f) Using `vars::causality()`, test whether Germany Granger-causes the rest of the system. Report the null hypothesis, the test statistic, and the p-value. What do you conclude at the 5% level of significance? How

about the 10% level of significance? What is the implication of your findings for the predictive content of Germany's growth rate for the rest of the Euro area?

Null hypothesis: Germany does not Granger-cause any of the rest of the system Test-stat = 2 p-value = 0.004

At both 5% and 10% significance level we reject the null. We conclude that Germany does Granger-cause the rest of the system. We conclude this due to the p-value of .004 being lower than both .05 and .1. The implications of this are that Germany is a good indicator of future values of growth rates fluctuations for France and Italy.

```
germany_causality <- causality(var_model_p2, cause = "g_ger")
print(germany_causality$Granger)
```

```
##
## Granger causality H0: g_ger do not Granger-cause g_fr g_it
##
## data: VAR object var_model_p2
## F-Test = 2, df1 = 4, df2 = 177, p-value = 0.04
```

Part IV: Impulse Response Functions

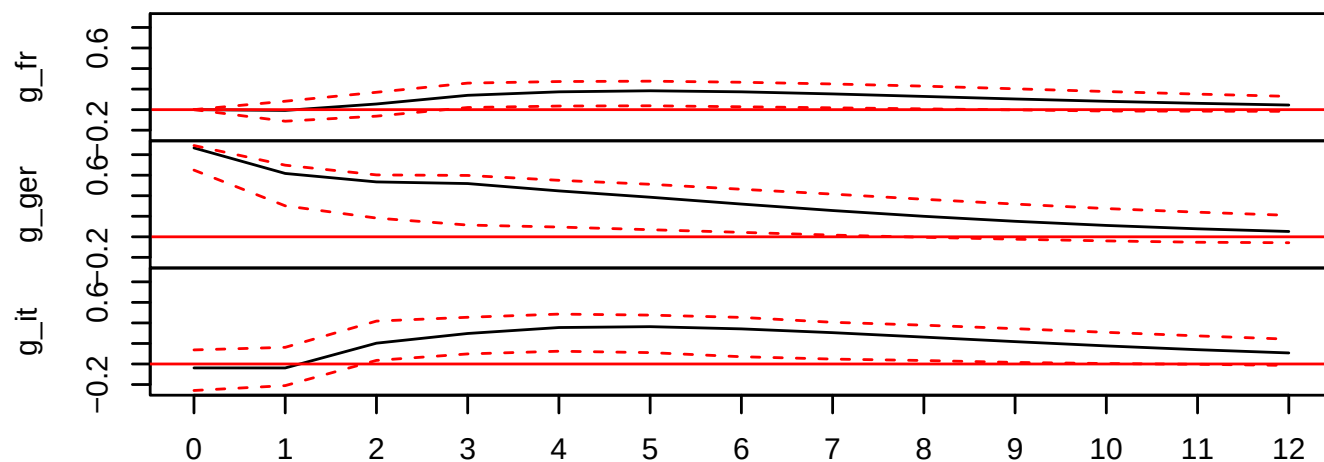
- (g) Compute and plot the orthogonalized IRFs for a one-unit shock to German GDP growth (`g_ger`), tracing the response of all three European series over 12 quarters. Use Cholesky identification with the ordering (France, Germany, Italy) and bootstrap confidence intervals (`boot = TRUE`, `ci = 0.95`). Are the spillovers from Germany to France and Italy positive and statistically significant? At what horizon do they peak?

The spillovers are positive and they are also statistically significant. France peaks out at either 4 or 5. Germany peaks in quarter 0 while Italy peaks in quarter 4-5.

```
irf_ger <- irf(var_model_p2, impulse = "g_ger",
               response = c("g_fr", "g_ger", "g_it"),
               n.ahead = 12, ortho = TRUE, boot = TRUE, ci = 0.95)

plot(irf_ger)
```

Orthogonal Impulse Response from g_ger



95 % Bootstrap CI, 100 runs

- (h) The Cholesky ordering places France first, implying it does not respond contemporaneously to shocks from Germany or Italy within a quarter. Re-estimate the IRFs using the alternative ordering (Germany, France, Italy) and briefly compare the results. How sensitive are your conclusions to this new ordering? **Note: You will need to re-estimate the VAR with the new ordering of the endogenous variables to do this. The exogenous variables remain the same.**

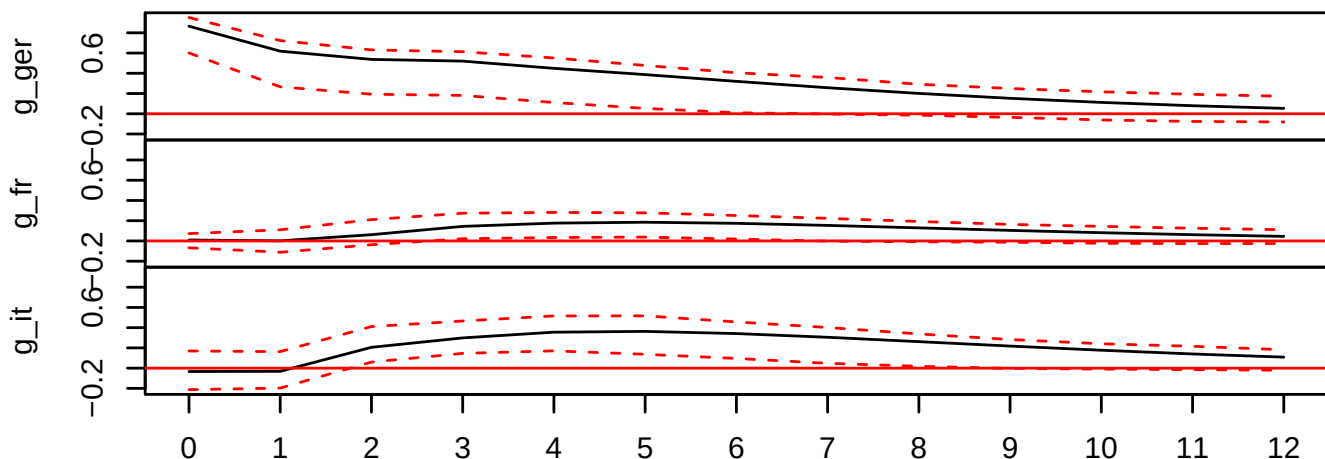
The Italian ones do not have reaction again until quarter 1 while France starts to go up around that time as well. Germany still has a very strong and positive response at Quarter 0 but decays steadily. The shapes of the graph virtually do not change too much and is pretty similar, signaling that the ordering does not matter much to the model. The conclusions stay the same as the previous ordering so I would say that the conclusions are not sensitive to the new ordering. Germany still has significant and positive spillover to both French and Italy.

```
y_train_alt <- train_data %>%
as.data.frame() %>%
dplyr::select(g_ger, g_fr, g_it)

var_model_alt <- VAR(y_train_alt, p = 2, type = "const", exogen = exog_train)

irf_alt <- irf(var_model_alt, impulse = "g_ger",
               response = c("g_ger", "g_fr", "g_it"),
               n.ahead = 12, ortho = TRUE, boot = TRUE, ci = 0.95)
plot(irf_alt)
```

Orthogonal Impulse Response from g_ger



95 % Bootstrap CI, 100 runs

Part V: Forecast Error Variance Decomposition

- (i) Compute the forecast error variance decomposition (FEVD) for the original VAR specification (France, Germany, Italy) over a 12-quarter horizon. Using the `fevd_make_table` helper function from the class notes, create a table summarizing the contributions of each shock at 1-, 4-, 8-, and 12-quarter ahead horizons.

Briefly summarize your findings. Focus on the relative contributions to own and cross-variable forecast error variances. What do your results imply about the importance of German growth shocks for the Euro area business cycles?

Germany seems to have a higher impact on France and Italy while seemingly not being affected by their actions. In the first quarter most of each countries growth is found upon their own domestic shocks. Germany in the last quarter like 12, seems to be a high explainer of shocks in France in Italy.

```
make_fevd_table_simple <- function(fevd_element, h = c(1, 4, 8, 12),
                                   row_names = c("France Shock", "Germany Shock", "Italy Shock")) {
  tab <- t(data.frame(fevd_element)[h, ])
  rownames(tab) <- row_names
  colnames(tab) <- paste0("H-", h)
  return(round(tab * 100, 2))
}
```

```
fevd_res <- fevd(var_model_p2, n.ahead = 12)
```

```
print("FEVD for France (g_fr):")
```

```
## [1] "FEVD for France (g_fr):"
```

```
make_fevd_table_simple(fevd_res$g_fr)
```

```
##           H-1   H-4   H-8 H-12
## France Shock 100 89.02 70.78 65.9
## Germany Shock  0  5.22 24.49 28.7
## Italy Shock    0  5.76  4.73  5.4
```

```
print("FEVD for Germany (g_ger):")
```

```
## [1] "FEVD for Germany (g_ger):"
```

```
make_fevd_table_simple(fevd_res$g_ger)
```

```
##           H-1   H-4   H-8 H-12
## France Shock  0.06  1.74  2.13  2.51
## Germany Shock 99.94 96.20 94.98 93.50
## Italy Shock    0.00  2.06  2.89  3.99
```

```
print("FEVD for Italy (g_it):")
```

```
## [1] "FEVD for Italy (g_it):"
```

```
make_fevd_table_simple(fevd_res$g_it)
```

```
##           H-1   H-4   H-8 H-12
## France Shock  6.32 12.3  8.36  7.59
## Germany Shock  0.28 12.6 38.91 44.61
## Italy Shock    93.40 75.1 52.73 47.80
```

Forecasting

- (j) Mawuena now feels comfortable using the VAR to produce forecasts for the three European growth rates for the next 3-years.

Use your VAR with the ordering (France, Germany, Italy) to produce dynamic forecasts for all three series for the next 3-years (`n.ahead = 12`). You will need to supply the actual values of the exogenous variables (`g_us` and the recession dummies) over the forecast period of interest using the `dumvar` argument in the `predict()` function. **Note that the future values supplied should correspond to observations 73-84 of the exogenous variables in the dataset.**

Using the `fanchart()` function, plot the forecasts for all three series. Briefly comment on the forecasted trajectories of the three series and the associated uncertainty.

They all do a similar path of mean reversion and smooth out trajectories over the forecasted period. The fan also spreads out considerably over time relative to the beginning of the forecasted period. You could say that this model is good for short term forecasting leaving more uncertainty in later quarters of forecasting.

```
exog_future <- euro_growth_raw[73:84, ] %>%
dplyr::select(g_us, starts_with("d_")) %>%
as.matrix()

var_forecasts <- predict(var_model_p2, n.ahead = 12, dumvar = exog_future)

fanchart(var_forecasts, colors = c("grey70", "grey80", "grey90"))
```

